Security Watch

IM With Caution

The Problem

A vulnerability has been reported in Yahoo! Messenger. This can potentially be exploited to compromise a user's system. The vulnerability is reported in versions obtained prior to 2 Nov 2006. It is caused due to an error in an ActiveX control, and can be exploited to cause a buffer overflow.

The Solution

Upgrade to the latest version, available at **http://messenger.yahoo.com**.

Firefox Vulnerable

The Problem

Security firm Secunia has given its "highly critical" rating to multiple vulnerabilities that have been reported in Mozilla Firefox. These can be exploited to gain knowledge of certain information and potentially compromise a user's system. The vulnerabilities are reported in all Firefox 1.x versions, as well as in Firefox 2.

The Solution

Mozilla released Firefox version 2.0.0.1 pretty soon after the vulnerabilities were reported. Just "check for updates" from the Help menu in Firefox—it's a small download. If that doesn't work—if it says "update failed"—then head to www.getfirefox.com and download the latest version.

If for some arcane reason you don't like Firefox 2, you should upgrade your other version to 1.5.0.9.

year under a pilot project, and the plan is to migrate fully by 2010. Where Linux goes, OpenOffice can't be far behind: the police unit has also shifted its users to OpenOffice and Firefox.

The bottomline? A Gendarme, speaking on condition of anonymity, said that while he was optimistic about Linux, he was not too happy with OpenOffice. He said he "missed" MS Office. "OpenOffice is complicated. It is atrocious," he said. (Something must have been lost in translation, we're sure. Unless he was speaking less-than-perfect English.) "We save money but the advantages of its use are not terribly clear." A mixed statement if ever there was one, but we get the general idea.

LANDMARK ACHIEVEMENT

Fujitsu Joins The Race

Seagate won the race with perpendicular recording of hard disks, of which you've doubtless heard. They produced the first 750 GB disk using that technology. Now, we mentioned in our December 2006 issue that they're already talking about HAMR to increase areal densities. Fujitsu recently achieved a landmark in this field of research.

HAMR (Heat-Assisted Magnetic Recording), as it is called by Seagate, or TAR (Thermally Assisted Recording), as it is called by Hitachi GST and others, uses a two-fold principle to record at higher densities. The first idea is that bits cannot be packed closer together because when this is done, they tend to lose their proper magnetic orientation. If the material used were of higher coercivity—meaning that it is harder to demagnetise—then the bits can be cramped together more closely without fear of them changing magnetisation.

The second idea is that if a material has higher coercivity, it's also harder to write on to, and that's where the heat comes in—when the spot of data to be recorded on is heated by a laser, it becomes easier to record on.

Now for recording onto heated spots, areal densities (how closely-cramped the bits can be) would be high enough only if the spot were small enough. As you can see, making a smaller spot is plainly key to the process.

Now Fujitsu has demonstrated what it claims is the world's first optical element capable of producing a beam less than 100 nm across. The company says TAR will be key to the next generation of hard disks in terms of capacity.

The company says the optical element will be able to achieve a spot size of 88 nm by 60 nm, eventually enabling recording at a density of 1 terabit per square inch or higher. This is more than double the current record—421 gigabits per square inch—set by Seagate earlier this year.

Other companies such as Hitachi GST (Global Storage Technologies) are also working on TAR (or HAMR). They say they can get the grain, or spot, size down to 2 nm, where conventional recording could only manage 8.

The bottomline: Fujitsu has said that in conjunction with perpendicular recording, HAMR (or TAR) will enable drives with capacities ten times higher than what is possible now.

As for "when," the timeline for TAR is about 2010. Another race, one we're keenly looking forward to—and one that's spoken about much less than the gigahertz races or the core wars. For some

HOT

HAMR

Heat-Assisted Magnetic Recording, or TAR (Thermally Assisted Recording). The hottest thing happening in storage research labs. IBM has a working prototype. Expect all disk storage news in 2007 to be about HAMR!

NOT

Perpendicular Recording

Sorry, but we measure time in weeks and days, not years. We don't expect anything groundbreaking to happen on this front—just produce more disks and get it done with, don't talk about the technology any more!

One Silly Question "How Big Is The Internet?"

"As endless as the universe!"
Amit Babar

"Bigger than the ocean."
Mayur Gangar

"It's like the sky."
Pradnya Gurav

"As big as my brain..."
Prashant Sonpal

"It stretches from the north to the south pole!"
Sudipta Satpathy